

SOC Defensive CTF Machine

Blue Team Investigation Lab — Full Project Walkthrough

Internship Project Documentation

SOC Defensive CTF Machine

An interactive blue-team training application that simulates a Security Operations Center investigation workflow using fictional incidents, alerts, logs, and evidence.

Prepared by	Dharmik Khunt
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Data classification	Fictional / simulated data only

1. Project Overview

The SOC Defensive CTF Machine is an interactive blue-team cybersecurity training application. It simulates a Security Operations Center (SOC) investigation workflow using fictional incidents, alerts, logs, evidence, investigation questions, analyst actions, and hidden flags.

Its purpose is to let a beginner SOC analyst practice investigating realistic security scenarios without attacking or touching real systems.

Technology Summary

Frontend	React + Vite (JavaScript/JSX)
Backend	Node.js + Express
Data	JSON/JavaScript-based simulated SOC data
User interface	Plain CSS with Lucide icons
Deployment mode	Local internship/demo lab with fictional data
Primary workflow	Dashboard → Cases → Investigation → Flag; Alerts → Related Case → Investigation

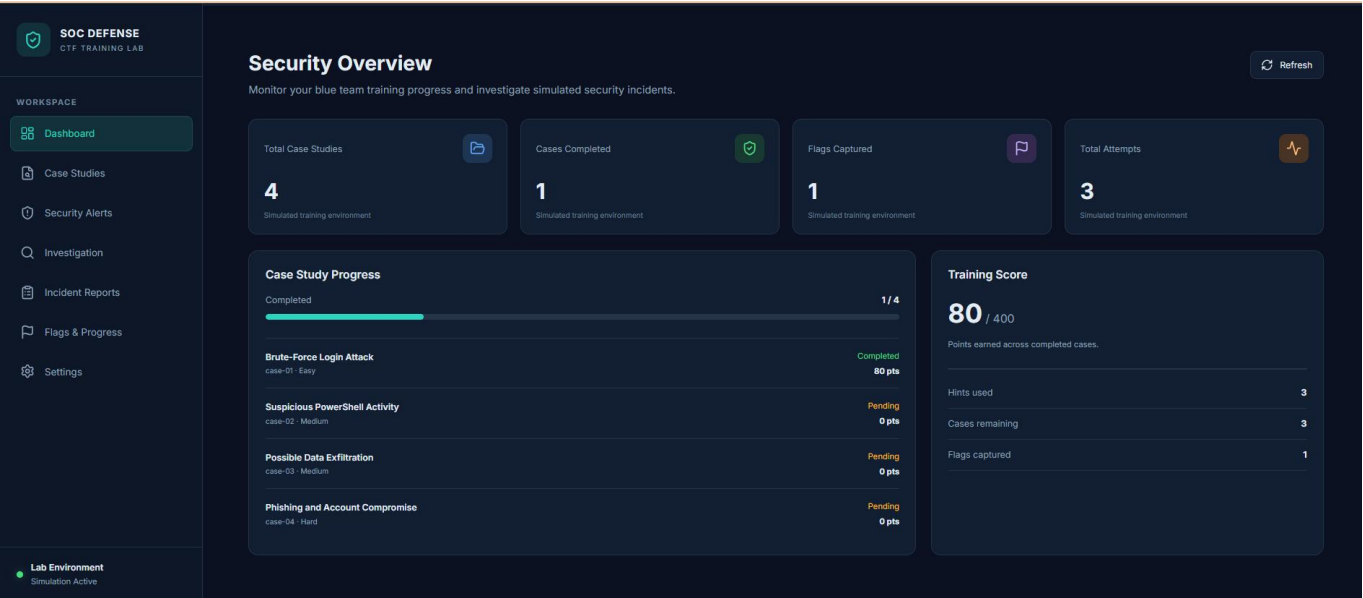


Figure 1: SOC dashboard with sidebar navigation, summary cards, and case progress.

2. Project Objectives

The project was designed to demonstrate the following SOC analyst skills:

- Reviewing and prioritizing security alerts
- Investigating simulated authentication and endpoint events
- Following an incident timeline
- Comparing suspicious activity with normal behavior
- Answering investigation questions using evidence
- Applying basic defensive response actions
- Tracking investigation progress and captured flags
- Producing an analyst-oriented report

3. Project Structure

The project is organized as a full-stack application, divided into a client (React frontend) and a server (Express backend).

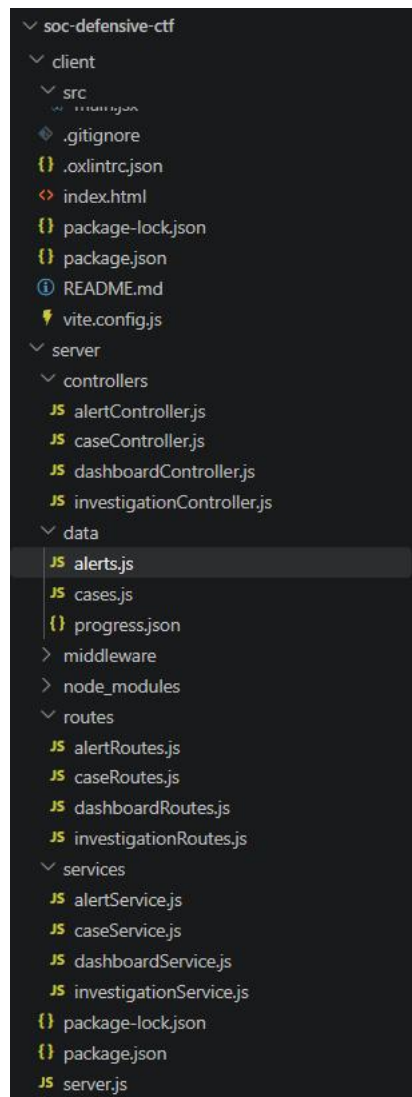


Figure 2: Project folder structure in VS Code, showing a structured full-stack application.

4. Backend Setup and API Layer

The backend is an Express server running locally on port 5000. It provides APIs for cases, investigations, dashboard statistics, alerts, progress tracking, hints, and alert status updates.

API Reference

Method	Endpoint	Description
GET	/api/health	Confirms that the backend is running
GET	/api/cases	Returns sanitized case information
GET	/api/cases/:id	Returns a single case
POST	/api/investigation/:caseId/submit	Checks analyst answers
GET	/api/investigation/:caseId/progress	Returns investigation progress
POST	/api/investigation/:caseId/hint	Provides a limited hint
GET	/api/dashboard?userId=analyst-01	Returns dashboard statistics
GET	/api/alerts	Returns simulated SOC alerts
PATCH	/api/alerts/:id/status	Sets an alert to Acknowledged or Resolved

```
PS C:\Users\ADMIN\soc-defensive-ctf\soc-defensive-ctf> cd server
PS C:\Users\ADMIN\soc-defensive-ctf\soc-defensive-ctf\server> node server.js
SOC CTF backend running at http://localhost:5000
█
```

Figure 3: Health endpoint confirming the Node.js/Express backend is running.

5. Case Studies

The machine contains four defensive SOC investigation cases.

Case 01 — Brute-Force Login Attack

The analyst investigates repeated failed logins followed by a successful login, and must identify the targeted account, source IP, failed attempt count, success time, and an appropriate response.

Case 02 — Suspicious PowerShell Activity

The analyst investigates a simulated process event in which PowerShell is launched by Microsoft Word. The focus is on the endpoint, process, parent process, suspicious indicator, and defensive response.

Case 03 — Possible Data Exfiltration

The analyst compares an unusually large outbound transfer with the workstation's normal traffic baseline, then evaluates the destination and business purpose.

Case 04 — Phishing and Account Compromise

The analyst follows a suspicious email, a link click, and a subsequent unusual login to determine whether the same user and timeline are connected.

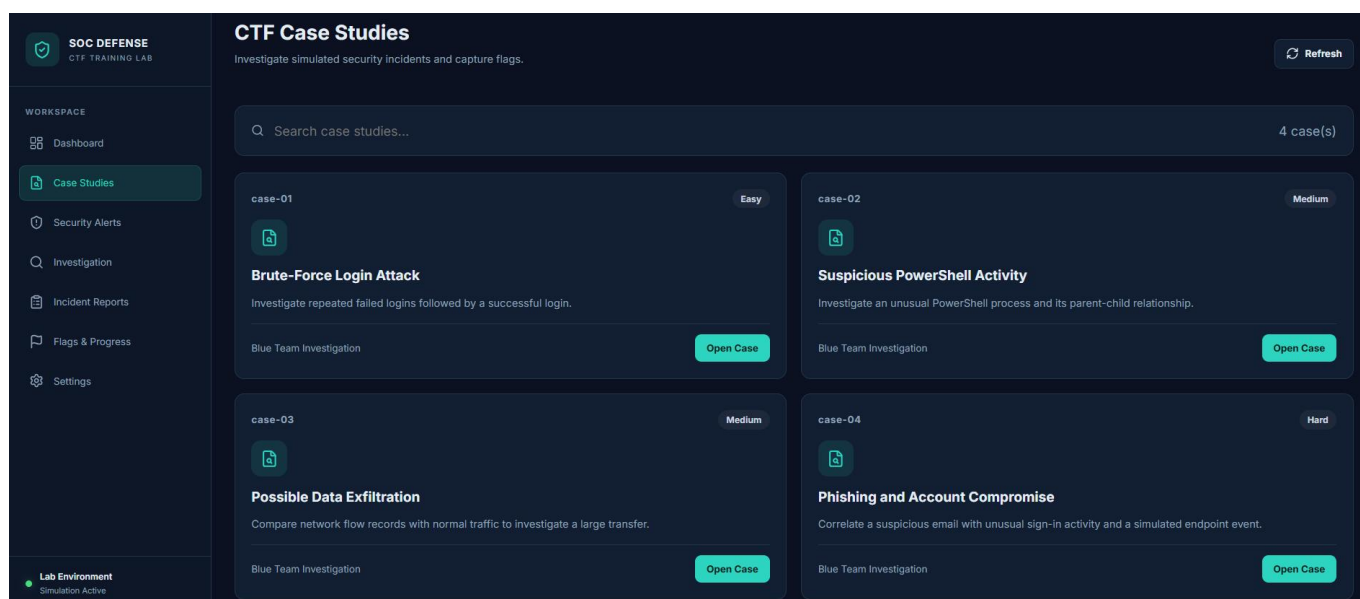


Figure 4: Cases page: the entry point to the four blue-team challenges.

6. Case 01 Investigation Walkthrough

Case 01 demonstrates the core CTF investigation process. The analyst reviews evidence and logs, identifies the repeated failed login attempts, notices the later successful login, and answers five investigation questions.

Key Simulated Evidence

Indicator	Value
Targeted account	alex.morgan
Source IP	198.51.100.24
Failed attempts	5
Successful login	09:13:04
Detection	BRUTE_FORCE_SIM
MITRE ATT&CK mapping	T1110 (Brute Force)

Note: All IP addresses, users, and events in this project are fictional and simulated.

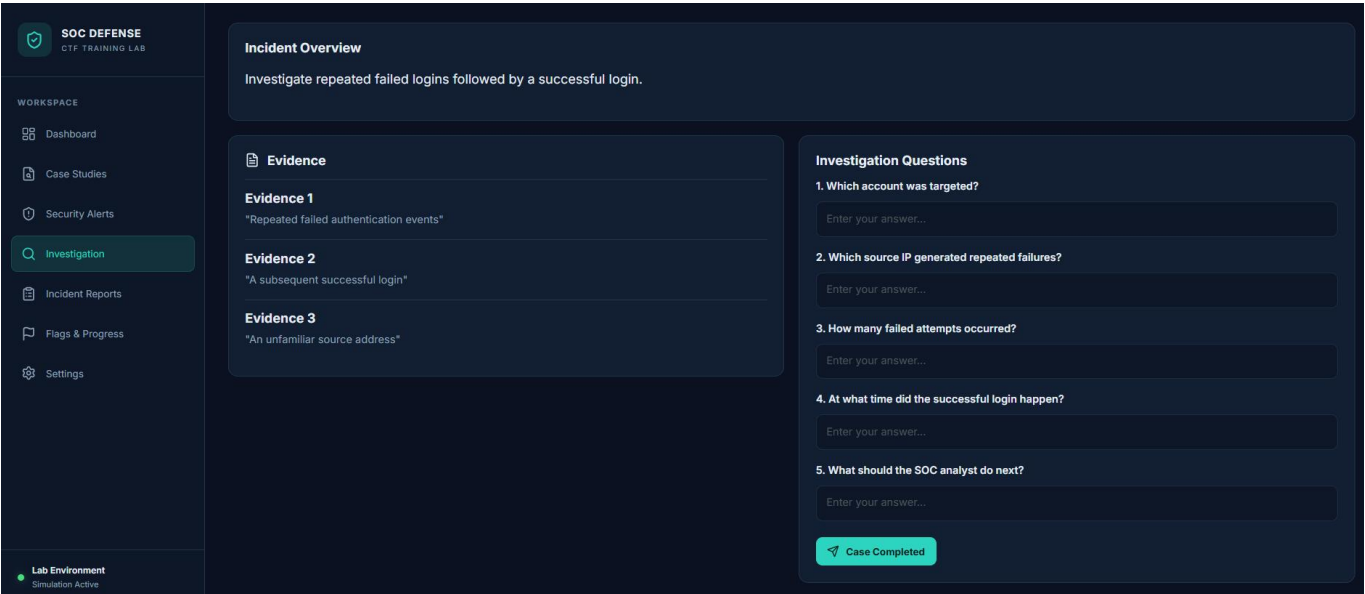


Figure 5: Case 01 investigation view: evidence must be analyzed, not simply clicked through.

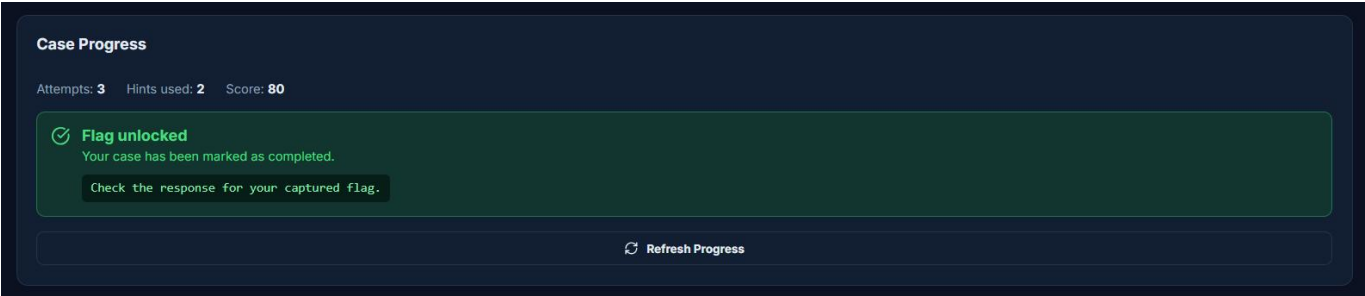


Figure 6: Case 01 answer submission and result.

7. Hint and Progress System

The investigation module includes a controlled hint mechanism. Each case can provide a limited number of hints, and the scoring system reduces the available score when hints are used. Progress tracking records attempts, hints used, completion state, and score.

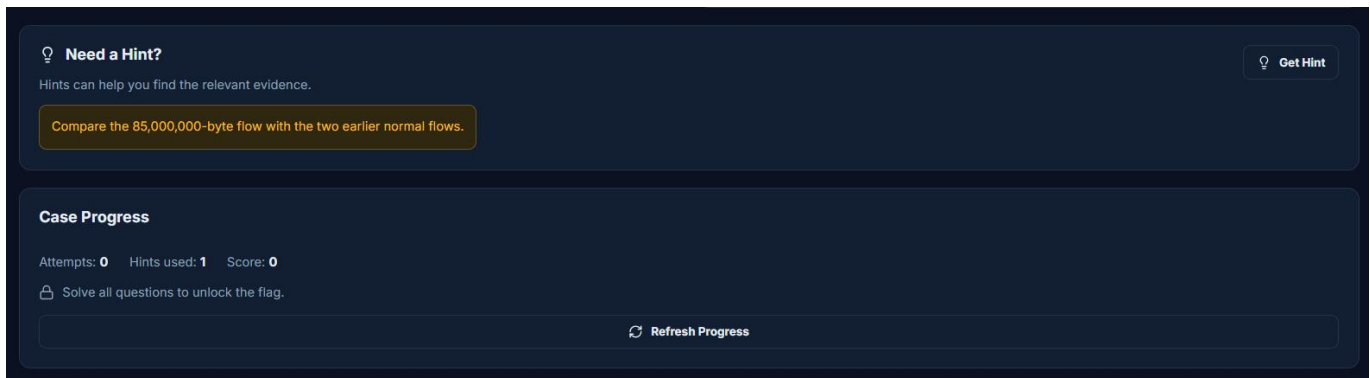


Figure 7: Progress panel and hint display, showing the built-in learning mechanism.

8. Flag Unlock Mechanism

The flag is not exposed in the normal case data. The backend keeps the answer key, flag, and hint protected from the standard case response. When the analyst submits all required answers correctly, the case is marked completed and the flag is returned.

For Case 01, the simulated flag is **CTF{BRUTE_FORCE_DETECTED}**.

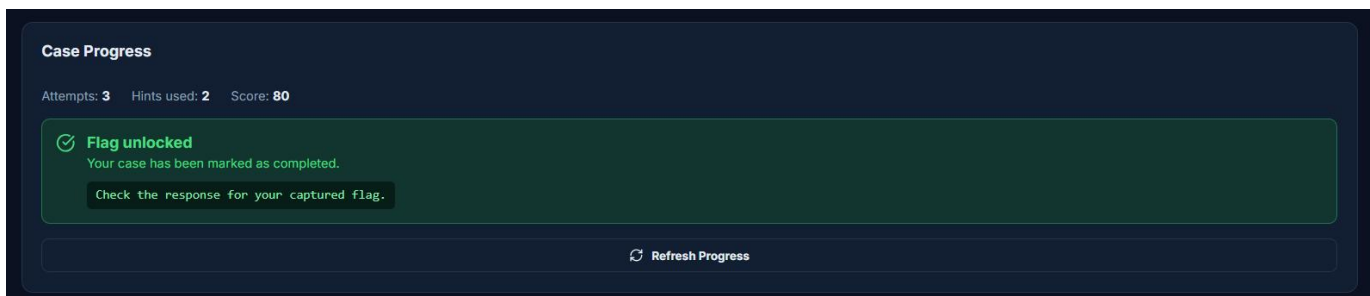


Figure 8: Flag unlocked after successful completion of Case 01.

9. Security Alerts Module

The Alerts page simulates a SOC alert queue. Analysts can search alerts, filter by severity, open alert details, and perform response actions. Key capabilities include:

- Alert listing with severity indicators
- Search functionality
- Severity filtering
- Alert detail modal
- Relationship between each alert and its investigation case
- Acknowledge and Resolve actions

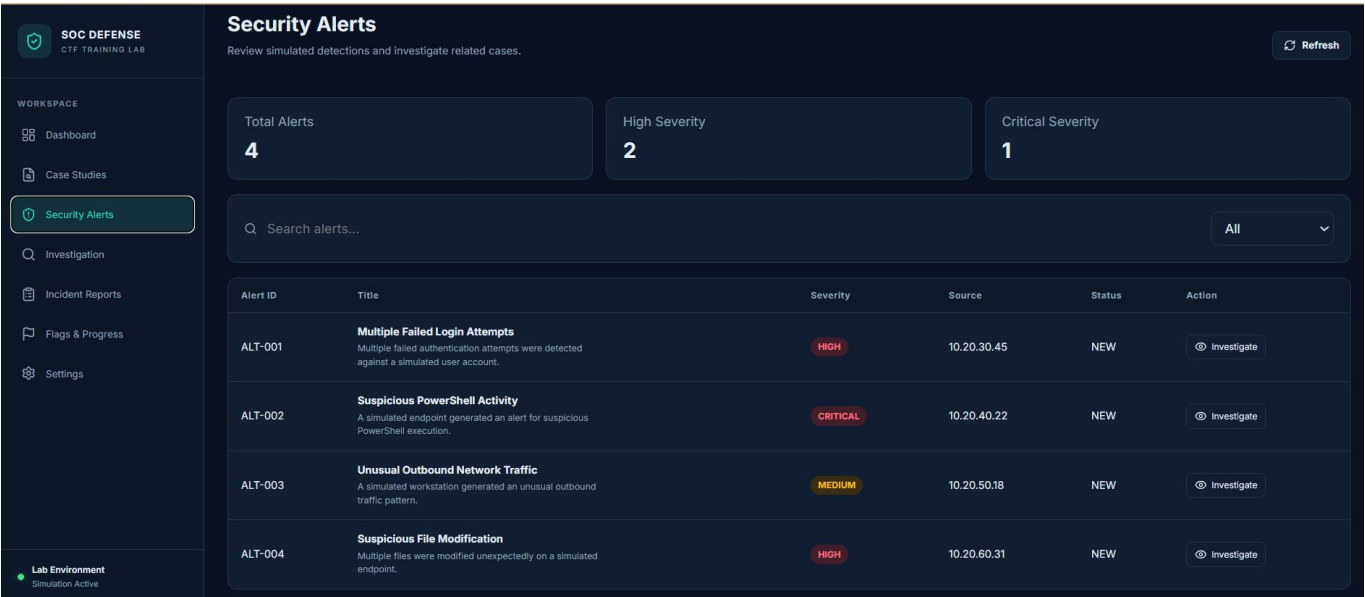
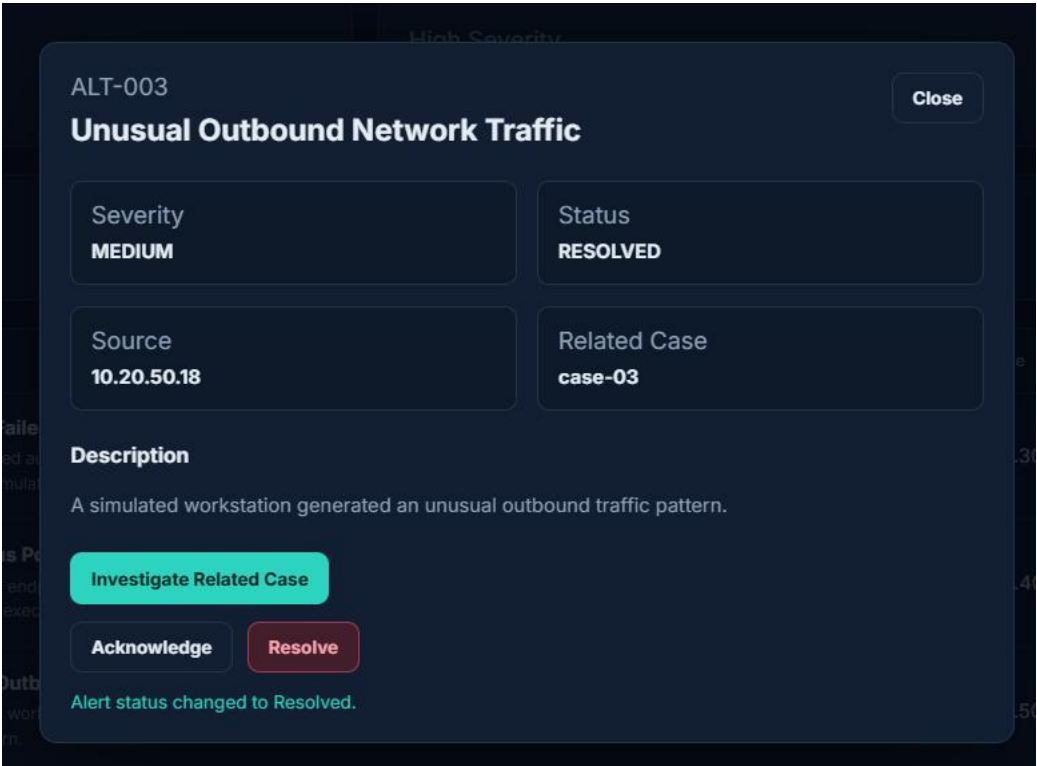


Figure 9: Security Alerts page representing the SOC alert-monitoring workflow.



10. Alert-to-Investigation Integration

Each simulated alert can be associated with a case through a caseId field. For example, ALT-001 is linked to case-01. From the alert detail view, the analyst selects **Investigate Related Case** to open the relevant investigation directly.

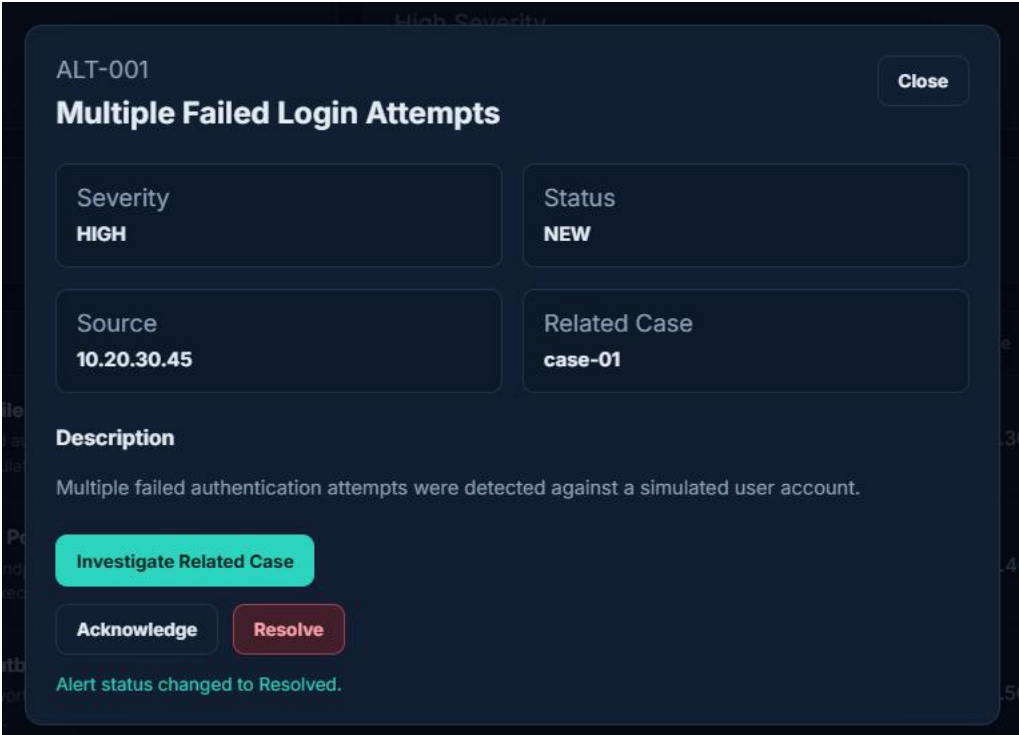


Figure 11: Alert triage connected to deeper investigation.

11. Alert Status Management

The alert response flow supports three statuses: New, Acknowledged, and Resolved. When an analyst changes an alert’s status, the frontend sends a PATCH request to the backend.

Demo limitation: In the current implementation, status changes are stored in memory and reset when the server restarts.

ALT-002	Suspicious PowerShell Activity A simulated endpoint generated an alert for suspicious PowerShell execution.	CRITICAL	10.20.40.22	ACKNOWLEDGED	Investigate
ALT-003	Unusual Outbound Network Traffic A simulated workstation generated an unusual outbound traffic pattern.	MEDIUM	10.20.50.18	RESOLVED	Investigate

Figure 12: Updated alert status, demonstrating a basic SOC response lifecycle.

12. Flags Page

The Flags page provides a challenge-completion overview. It lists completed and pending cases without exposing hidden flags unnecessarily.

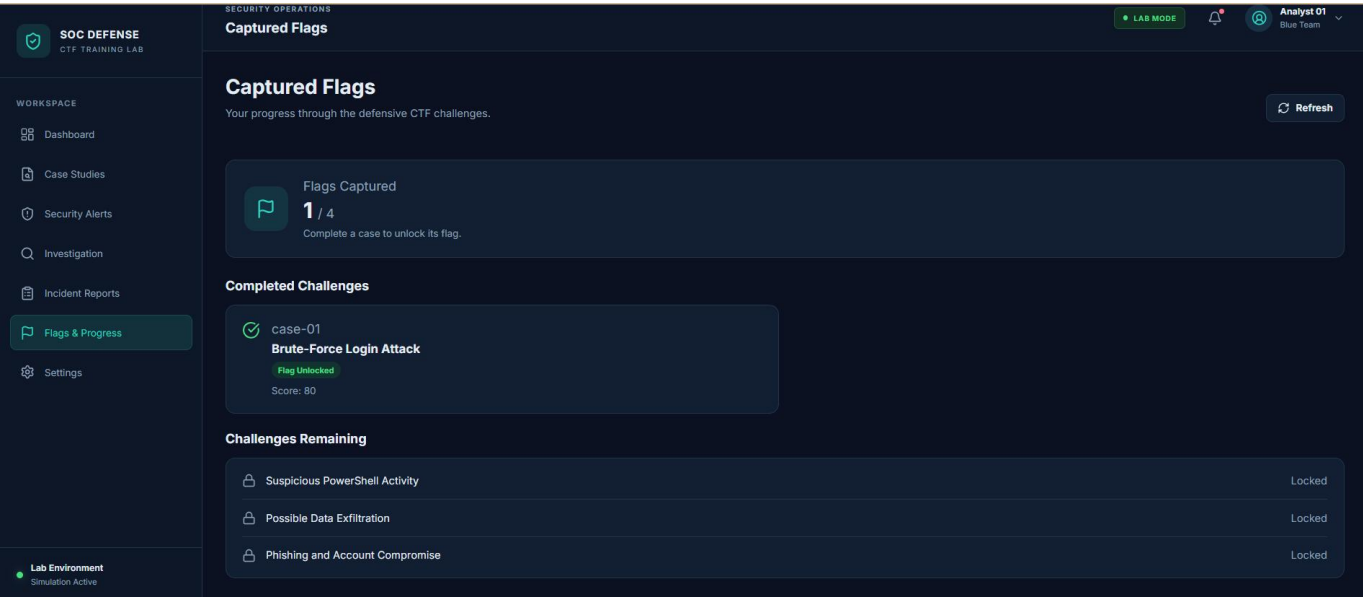


Figure 13: Flags overview tracking challenge completion across all cases.

13. Reports Page

The Reports page converts investigation progress into a concise, analyst-oriented summary. It includes:

- Completion percentage and progress visualization
- Total attempts, hints used, and score
- Per-case performance
- Refresh, print, and text export functionality

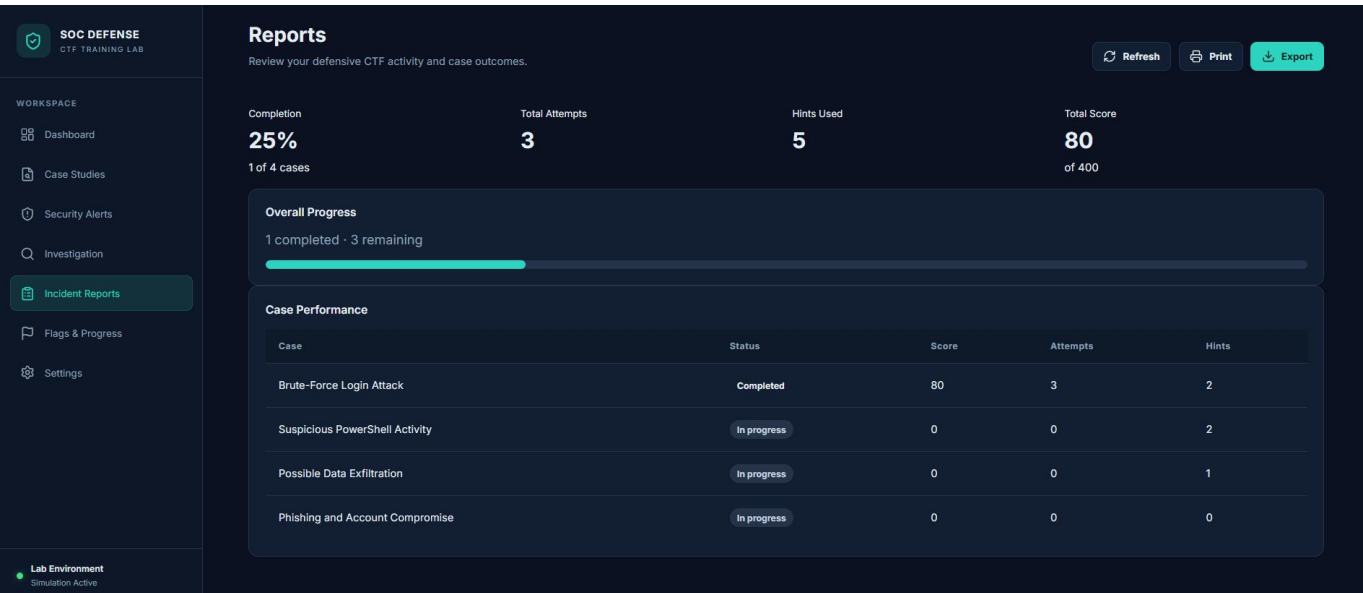


Figure 14: Reports dashboard summarizing investigation activity.

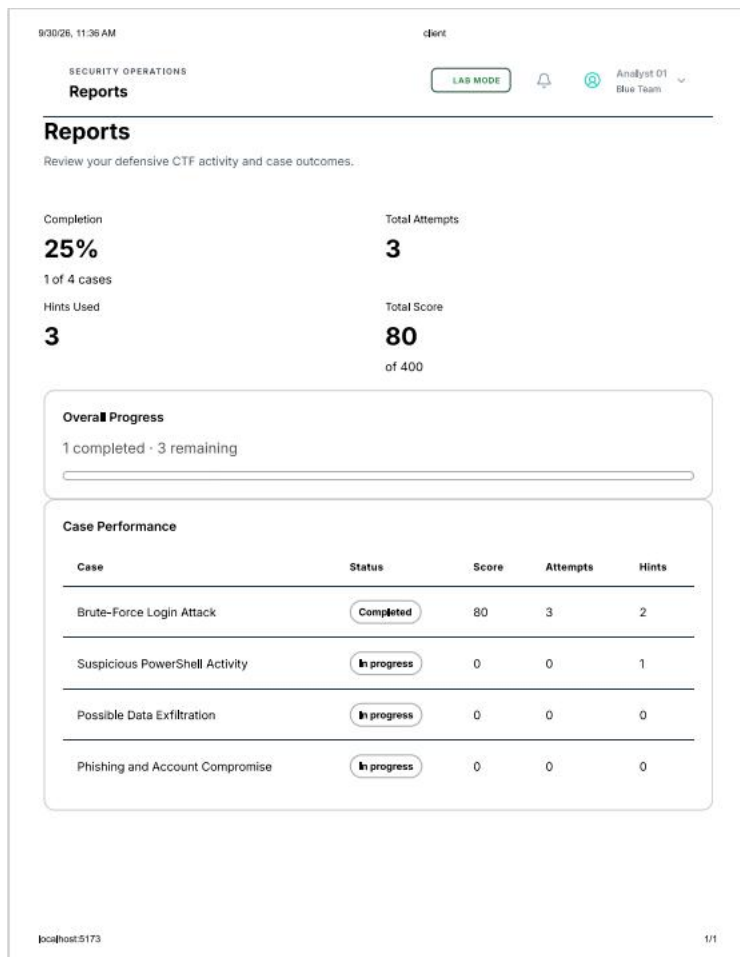


Figure 15: Report print and export controls.

14. Full End-to-End Workflow

The final integrated workflow proceeds as follows:

1. Start the Express backend on port 5000.
2. Start the React/Vite frontend on port 5173.
3. Open the Dashboard and review SOC statistics.
4. Open Cases and select a case.
5. Review the evidence, logs, and event timeline.
6. Answer the investigation questions.
7. Use a hint if required.
8. Submit the investigation.
9. Receive the flag after successful completion.
10. Open Alerts and review a related security alert.
11. Open the related investigation from the alert.
12. Acknowledge or resolve the alert.
13. Review the updated Flags and Reports pages.

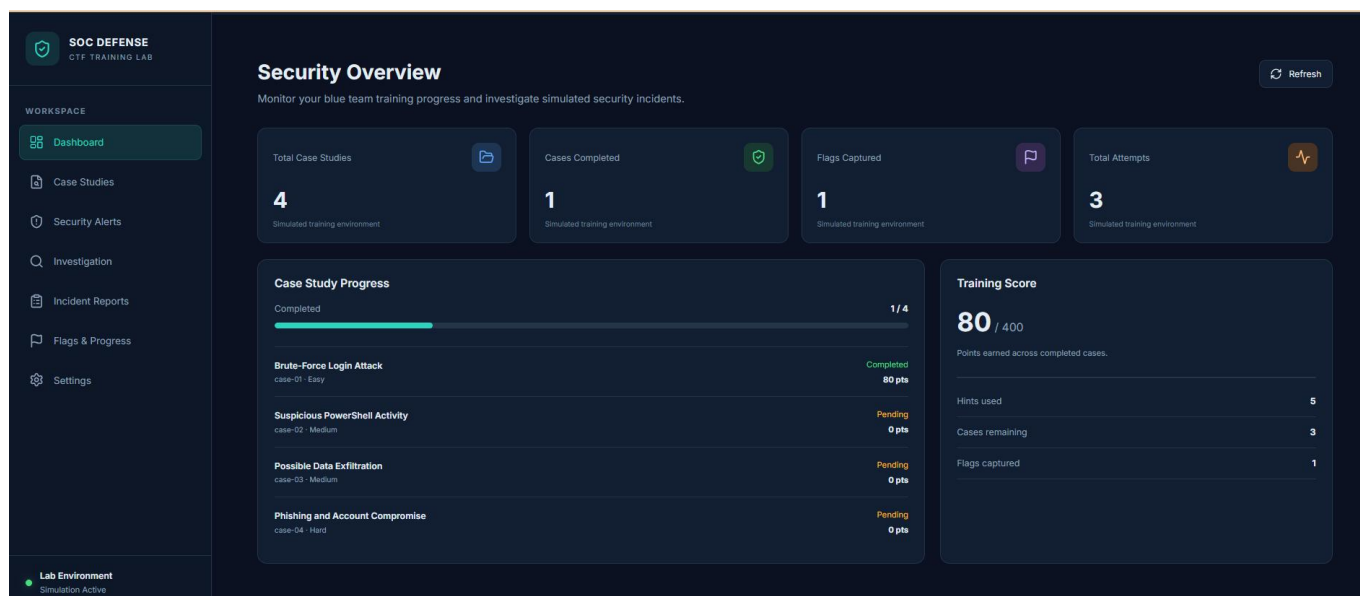


Figure 16: Dashboard after completing a fictional and used only for simulation.